Electro Group and Electrotech Services

End to end company case study

Department functions, operating workflows, application features and data requirements

Prepared for company process mapping and operations planning

15 October 2026

The existing project provides a starting point for engineering calculations, calibration reporting and employment letters. The proposed operating model connects these functions to customer orders, material, production, quality, finance, service and employee processes.

This package covers 24 department functions, 192 feature requirements, 488 data field entries, six workflows with 48 steps, 20 control requirements and 72 proposed performance measures.

The companion Excel register contains the complete field inventory, all retrieved wire rows, engineering reference options, catalogue entries, process steps and outstanding data requests.

Evidence labels distinguish source code, live spreadsheet data, public company statements and proposed capabilities. Financial and operating outcomes require company records.

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1 Executive case and scope

The company needs a joined operating record from customer requirement to accepted product or service and financial closure. The existing app already handles selected technical calculations and document preparation. The next operating improvement is to connect those outputs to approved orders, jobs, material transactions, reviewed releases and accountable owners.

This is an operational case study and requirements baseline. It includes the current project and a proposed complete functional model. The 24 functions are a coverage structure for validation; they are not presented as the company's verified organisation chart. One person or team may perform several functions.

Evidence and interpretation

Label	Meaning	Use in this study
Present in source	Observed in local app files	Confirms implemented code; live execution and adoption not established
Live spreadsheet data	Read from the connected sheet on the study date	Confirms 32 wire records and the two visible sheet tabs
Public company statement	Published on the matching company websites	Supports business scope; independent performance and certification not established
Proposed	Recommended process or capability	Requires process owner validation and implementation
Not supplied	No underlying internal records available in this study	No invented revenue, staff count, output, stock, cost saving or ROI

Current evidence and practical limits

The reviewed local files are the interface, application logic, spreadsheet backend, setup guide, service worker and application manifest. The connected workbook is titled Industrial Heater Coil Wire Database. Its metadata shows only nichrome_wires and kanthal_d_wires. All six relevant columns were read across the full 1000 row grid for each visible tab. There are 23 and 9 populated data rows respectively. [L01-L06, G01]

The code supports calibration_reports, performance_reports, offer_letters and appointment_letters. Those tabs were not visible in the connected workbook on the study date. This does not prove no reports or letters exist: browser local storage, another workbook or a different deployment may hold records. Operational document counts and historical transaction data remain unverified.

No profit improvement or time saving is claimed as achieved. The measurement section defines how to collect a baseline and verify results after implementation.

2 Company profile and business context

Verified study identity

The user confirmed Electro Group and Electrotech Services as the study scope. The project uses Electro Group of Industries branding for the operations interface and Electrotech Services branding for document workflows. The matching public website uses the name Electro Tech Services. [L01-L03, S01]

Item	Company published detail	Evidence
Business address	Plot No. Adj. S.M. 180, Industrial Estate Baikampady, New Mangalore 575011, Karnataka	S04
Website	https://electrotechind.in/	S01
Public business telephone	0824 2406417; +91 7022259317; +91 9141347017	S04
Public business email	electrotechservices97@gmail.com	S04
Service history statement	Website states servicing since 1995; legal incorporation date not established	S03
Associated heating business	The websites link to one another; legal ownership and group structure need confirmation	S12 S13
Associated address	Electro Heat Systems, D-123 Industrial Estate, Baikampady, New Mangalore 575011	S12
Associated history statement	Electro Heat Systems states manufacturing since 1998	S13

How the company creates value

The published business scope combines engineering and manufacture of heating and packaging equipment with repair, calibration and application specific machine work. This supports several operating patterns: custom design and manufacture, equipment supply, consumable supply, workshop repair, on site service and calibration. Actual revenue mix, order volumes and capacity are not available. [S01-S03]

The operating model therefore needs different job types. A stocked film sale requires item, quantity, price and dispatch controls. A custom oven needs approved design revisions, purchased components, fabrication, panel integration and customer acceptance. A calibration job needs instrument identity, standard evidence, raw readings and reviewed certificate issue. These differences drive the proposed workflows.

Identity and certification evidence to obtain

Confirm legal names, registration documents, sites, entity relationships, tax registrations, signatory authority, exact accreditation or product certification scope, and current certificates before configuring documents or making compliance claims. Website statements about traceability and testing do not by themselves establish a company accreditation or a licence covering every product.

3 Product and service portfolio

The following catalogue families were visible in the public sources. Catalogue entries can overlap across categories and brands; the count is not a deduplicated stock keeping unit count. Current price, stock, model variants, capacity, lead time, discontinued status and actual sales history were not supplied.

Family	Visible entries	Examples and evidence
Packaging machinery	12	Sealing, filling, coding and shrinking equipment. The category reports 14 results; page two was unavailable. S05
Plastic processing	3	Hopper drying/loading and loop handle equipment. S06
Heating equipment	7	Ovens, furnaces, generators, hot plates and related equipment. S07
Controls and panels	3	Timers/counters, panel boxes and temperature controllers. S08
Sensors	1	One thermocouple and sensor category entry. S09
Packaging materials	3	Shrink film, strapping and wrapping film. S10
Custom machines	6	Stirring, coding, sealing, industrial oven and conveyor entries. S11
Associated heating catalogue	15	Band, cartridge, tubular, infrared and other heater families with controls and sensors. S14

Service scope

Electrotech publicly describes repair, replacement and maintenance for electrical ovens, sealing machines and control panels. Its service page also describes temperature controller calibration and traceability to NABL certification. The study does not treat that wording as independent evidence of laboratory accreditation. Obtain the reference certificates, applicable scope, method and issuer authorisation for actual jobs. [S03]

Electro Heat Systems describes testing activities including electrical breakdown, insulation resistance, resistance and dimensions. These are published company statements, and the precise product plan, method, equipment and acceptance limits must be validated before a digital inspection workflow is released. [S13]

Portfolio completeness actions

The workbook contains all 50 catalogue entries retrieved in this study. Request the current complete company product master, including candle machinery details, the two inaccessible packaging entries, SKU identifiers, approved technical specifications and pricing. The candle category appears in navigation but its product page was unavailable. No item names were invented for the inaccessible entries.

Assign business line and order type independently. An oven can create a manufacturing order, a commissioning project and later a service ticket. Each remains linked to the same equipment serial while keeping its own scope, acceptance and commercial records.

4 Existing application capabilities

The current project is a browser application built from HTML, CSS and JavaScript with a Google Apps Script backend and Google Sheets storage. The manifest and service worker support an installable app shell. This section reports source observations, not an end to end live application acceptance test. [L01-L06]

Module	Capabilities observed	Boundary
Overview	Calibration total, pass, due soon, heater runs, offer, appointment and performance counts	These are app metrics, not a reconciled enterprise dashboard
Heater calculator	Material database, electrical and geometry calculations, automatic and manual inputs, result selection	No complete released design history or job costing record observed
Calibration	Editable certificate, readings, customer reuse, numbering, due date, register, print and CSV	Independent issue approval and immutable revision control not established
Performance	Calibration linking, shared context/sequence, Celsius or percentage error, separate register and export	Requires valid linked calibration and complete readings
Offer letters	Editable placeholders, live A4 preview, confirmation, register and export	Employment offer workflow; not payroll processing
Appointments	Four page document, compensation annexure, issue/register/export	Entered gross values and document issue do not establish signed acceptance
Storage	Spreadsheet calls, local record storage and fallback messages	Live workbook currently shows only wire tabs
Installation	Browser install prompts and same origin app asset caching	Offline shared data and durable background synchronisation not established

Existing data organisation

Wire tables use id, swg, thickness, ohm, minw and maxw. Four document tables have searchable index columns and store complete editable content in payload_json. Browser local storage keeps reports, customer reuse details, document numbering state and heater calculation count. A full heater calculation history is not established by the count alone. [L03 L04]

The backend validates administrator credentials for wire table writes. Source handlers for document saves do not use that wire administrator check. The setup guide calls out deployment access restrictions for shared internal records. The actual deployed access policy was not inspected. This is a concrete requirement for identity and record access review before wider use. [L04 L05]

5 Engineering logic and wire master findings

Calculation rules observed in source

Base resistance = voltage squared divided by wattage. Adjusted resistance = base resistance multiplied by one plus extra resistance percentage divided by 100. Wire length = adjusted resistance divided by the selected row resistance per metre. Turns = wire length in millimetres divided by pi times effective core diameter. Pitch = core winding length divided by turns. [L03]

Surface area in square centimetres = pi times heater outside diameter in millimetres times heater length in millimetres divided by 100. Watt density = wattage divided by that area. The code selects insulation thickness of 1.5 mm below 8 W/cm2, 1.7 mm from 8 to below 9, and 2.0 mm at 9 or above. These are application rules, not independently validated universal design limits. [L03]

The initial wire row is the first row whose inclusive wattage band contains the input. The code then evaluates subsequent rows. The pipe selection remains fixed while automatic core geometry is evaluated for each wire thickness. For SWG 22 to 32, the pitch pass threshold is 98 percent of twice the wire thickness. For SWG 33 to 45 it is twice the wire thickness. A released design requires engineering validation of these rules and operating conditions. [L03]

Actual data findings

Finding	Observed evidence	Required review
Repeated rows	Nichrome SWG 32, 36, 42 and 44 have exact repeated technical values with different IDs	Confirm intentional alternatives or remove ambiguity through controlled master revision
Different resistance for same gauge	Nichrome SWG 38 has 62.88 and 59.408; SWG 40 has 93.2 and 94 in the ohm column	Validate alloy, tolerance, supplier basis and intended selection key
Overlapping bands	For example Nichrome 900 to 1049 W matches both SWG 24 and 25 bands	Document precedence; current first match is order dependent
Uncovered Kanthal band	1591 to 1599 W is between the supplied 1590 maximum and 1600 minimum	Confirm intended band limits
Gauge acceptance coverage	Kanthal rows include SWG 20 and 21; source pitch acceptance branches start at SWG 22	Verify intended behavior for high wattage starting rows
Reference list meaning	Core and pipe values are hardcoded geometry references without quantities	Do not present them as actual available stock balances

6 Calibration and document behavior

Calibration result behavior

The current calibration calculation uses observed error = device reading minus standard input. Allowed absolute error = absolute standard input multiplied by the entered claimed error percentage divided by 100. Each completed row passes when its absolute error is within that limit with a small numeric comparison tolerance. Error display is rounded to two decimals. The overall report is Pass only when every reading passes; otherwise it is Review. [L03]

A zero standard input gives zero allowed error under this percentage of reading rule. The agreed tolerance basis must therefore be checked for each method. An absolute Celsius tolerance, percentage of full scale or uncertainty aware conformity rule cannot be assumed to match the current percentage of reading implementation.

Performance result behavior

Performance reports support absolute Celsius error or percentage error. Percentage error is observed minus reference divided by the absolute reference, multiplied by 100. A zero reference in percentage mode cannot be calculated. Completed reports use Pass or Fail; incomplete evaluation uses Pending. A linked calibration is required by the save validation. [L03 L04]

Numbering and due dates

Numbering uses browser sequence state and saved records, including manually entered sequence continuity. Backend save functions use locks for writing rows. This does not establish a globally unique controlled issue sequence across concurrent browsers. The proposed issue service must allocate unique numbers and retain the approved report revision. [L03 L04]

The default calibration due date is one calendar year after the calibration date less one day. For 15 October 2026 this gives 14 October 2027. This is a software default; the interval must follow the actual instrument, contract and approved calibration programme. [L03]

Required release additions

Before controlled issue, add reference validity and scope checks, approved method revision, competence authorisation, customer agreed decision rule, uncertainty where applicable, independent review, issued revision retention, controlled amendment and restricted export. The workbook separates these proposed features from the existing report builder.

Human resources document behavior

Offer and appointment workflows generate documents and maintain confirmed or issued records. Appointments have their own sequence and four page format with compensation information. Selection, accepted terms, signed acknowledgement, employee ID, attendance, payroll processing and statutory outputs require separate records and controls. Template clauses and default gross amounts must be approved by the company before use. [L02-L04]

7 Department coverage model

Each profile below states purpose, process, feature behavior, important records, handoffs and acceptance criteria. Feature IDs match the companion register. Proposed functions should be mapped to the real organisation and may share staff. All features are proposed unless marked Present in source.

Code	Function	Features
GOV	Management and strategy	8
MKT	Marketing and business development	8
SAL	Sales and customer relationship management	8
EST	Estimation and commercial costing	8
ENG	Engineering and research development	8
PPC	Production planning and control	8
WND	Heater production and winding	8
FAB	Mechanical fabrication and assembly	8
CTL	Electrical controls and automation	8
BUY	Procurement and vendor management	8
INV	Stores and inventory	8
QUA	Quality assurance and inspection	8
CAL	Calibration and metrology	8
PRJ	Projects installation and commissioning	8
SRV	Service repair and annual maintenance	8
LOG	Dispatch logistics and delivery	8
FIN	Finance accounts and treasury	8
HR	Human resources recruitment and onboarding	8
PAY	Attendance payroll and employee development	8
ADM	Administration and facilities	8
MNT	Maintenance and utilities	8
EHS	Environment health and safety	8
ITS	Information technology data and security	8
AUD	Compliance audit and document control	8

7 1 Management and strategy

Own business priorities, accountability, budgets and review decisions across the confirmed study scope.

Process Set plan, approve resources, review exceptions, assign actions and verify closure.

GOV-01 Operations dashboard *Present in source [L02 L03]*
Display calibration counts, passes, due soon, heater runs and HR/performance counts.

GOV-02 Business unit register
Separate legal entities, sites, business lines, authorised signatories and intercompany relationships.

GOV-03 Annual operating plan
Record sales, margin, production capacity and cash assumptions with accountable owners.

GOV-04 Budget approvals
Route spending and capital requests against a documented authority matrix.

GOV-05 Balanced performance review
Compare agreed targets with reconciled actuals and show period and data coverage.

GOV-06 Action register
Track decisions, owners, target dates, dependencies and closure evidence.

GOV-07 Risk register
Maintain operational, customer, supplier, safety and technology risks with mitigations.

GOV-08 Management review pack
Compile overdue commitments, quality trends, cash exposure and improvement decisions.

Records and fields entity id, site id, business line, period, budget version, target value, actual value, action owner, action due date, risk owner, decision reference.

Measures Action closure on time; Budget variance; Operating margin.

Handoffs Finance supplies reconciled results; all department leads own action closure.

Control Reviewers approve changes to targets and retain the previous plan version.

Acceptance A period review names a single owner for every unresolved action and links each displayed actual to its source.

7 2 Marketing and business development

Connect the published product and service portfolio to qualified enquiries and repeat business.

Process Maintain product content, record campaign response, qualify demand and hand accepted leads to sales.

MKT-01 Product content library
Maintain approved product families, specifications, application notes, images and revision owners.

MKT-02 Enquiry capture
Capture web, telephone, walk in, dealer and event enquiries with consent and source.

MKT-03 Campaign register
Record channel, audience, spend, response and qualified enquiry outcome.

MKT-04 Lead qualification
Classify application, location, estimated need and purchase timing before assignment.

MKT-05 Customer segmentation
Group accounts by application, service needs, geography and purchasing history.

MKT-06 Demonstration planning
Schedule equipment demonstrations and record requirements and follow up.

MKT-07 Partner register
Maintain dealer, referral and industry partner contacts and agreed responsibilities.

MKT-08 Content approval
Check claims, specifications and image rights before material is published.

Records and fields campaign id, lead id, lead source, product family, application, region, consent status, campaign cost, qualified date, assigned salesperson, content revision.

Measures Qualified lead conversion; Cost per qualified lead; Enquiry response time.

Handoffs Sales accepts qualified leads; engineering approves technical content; finance verifies campaign costs.

Control Use approved performance claims and keep a recorded basis for public product specifications.

Acceptance A new enquiry retains its source and owner through qualification and appears once in the sales handoff.

7 3 Sales and customer relationship management

Own customer requirements and commercial commitments from enquiry to accepted order.

Process Register enquiry, assess application, obtain technical costing, issue quotation, negotiate and confirm order.

SAL-01 Customer reuse *Present in source [L03]*
Reuse customer names and addresses across calibration and performance forms.

SAL-02 Customer account master
Use stable customer IDs, contacts, billing and delivery addresses and account status.

SAL-03 Opportunity pipeline
Track stage, next action, expected value, product line and loss reason.

SAL-04 Requirement capture
Record voltage, power, temperature, dimensions, environment, throughput and service scope as applicable.

SAL-05 Quotation follow up
Manage quotation revisions, expiry, negotiation and promised response dates.

SAL-06 Sales order review
Compare customer purchase order with accepted technical and commercial terms.

SAL-07 Credit and advance check
Obtain finance clearance or approved exception before order release.

SAL-08 Customer commitment history
Record changes to quantity, specification, price and delivery promise with approval.

Records and fields customer id, contact id, enquiry id, opportunity id, quotation id, quotation revision, customer po, order id, delivery commitment, payment terms, credit clearance.

Measures Quote win rate; Enquiry to quotation time; On time customer commitment.

Handoffs Engineering checks feasibility; estimation owns cost; planning receives approved order; finance clears credit.

Control Do not convert an unapproved specification or quotation into a production instruction.

Acceptance One accepted order links the customer requirement, quotation revision, purchase order and finance clearance.

7 4 Estimation and commercial costing

Produce traceable costs and prices for custom heaters, machines, service work and consumables.

Process Receive scope, choose costing basis, price material and labour, review margin and release quotation.

EST-01 Material costing
Calculate cost from approved bill of materials, units, scrap allowance and supplier price date.

EST-02 Labour costing
Estimate operation hours and approved labour rates by work centre or service skill.

EST-03 Overhead allocation
Apply an approved basis and keep overhead separate from direct cost.

EST-04 Outsourcing estimates
Include machining, treatment, bought out panels or other subcontract scope.

EST-05 Service estimate
Separate diagnosis, travel, spares, labour and exclusions.

EST-06 Margin review
Show net sales and cost basis before discount approval.

EST-07 Quotation revision control
Keep superseded prices and scope changes attached to the opportunity.

EST-08 Estimate versus actual
Compare completed job material, labour and rework with the accepted estimate.

Records and fields estimate id, revision, order id, bom revision, material cost, labour hours, labour rate, overhead basis, subcontract cost, quoted net value, discount approval.

Measures Estimate variance; Quoted gross margin; Quotation turnaround time.

Handoffs Procurement supplies prices; engineering supplies design; finance approves rates and commercial exceptions.

Control Use dated rate versions and disclose exclusions; approved price cannot be silently replaced.

Acceptance A reviewer can reconstruct a quotation from its quantities, rates, version and approved margin.

7 5 Engineering and research development

Turn customer needs into approved and repeatable designs across heater and equipment applications.

Process Review specification, calculate design, verify constraints, release drawing and revise through change control.

ENG-01 Heater electrical calculation *Present in source [L03]*
Calculate base resistance and adjusted resistance from wattage, voltage and extra resistance.

ENG-02 Material selection *Present in source [L02 L03]*
Choose Nichrome or Kanthal D and load the corresponding wire data.

ENG-03 Geometry and gauge selection *Present in source [L03]*
Evaluate pipe, insulation, core diameter, wire length, turns and pitch against implemented rules.

ENG-04 Manual design overrides *Present in source [L02 L03]*
Allow manual pipe and core inputs alongside automatic selections.

ENG-05 Wire master editing *Present in source [L03 L04]*
Validate numeric wire rows and require backend administrator validation for writes.

ENG-06 Design package release
Link calculation snapshot, drawing, bill of materials and approved revision to a job.

ENG-07 Engineering change control
Assess stock, work in progress, customer approval and cost impact before changing a released design.

ENG-08 Prototype validation
Record thermal, electrical, mechanical and application test evidence against an agreed plan.

Records and fields design id, design revision, material, wattage W, voltage V, heater od mm, heater length mm, core length mm, pipe od mm, pipe wall mm, core od mm, extra resistance pct, swg, wire ohm per m, wire length m, turns, pitch mm, watt density W cm2, approved by.

Measures Design release lead time; Design change after release; Prototype first pass rate.

Handoffs Sales supplies operating conditions; procurement confirms available material; planning uses released revision.

Control Application rules are company logic and require engineering validation for actual operating conditions.

Acceptance A released design reproduces its inputs, wire master version, calculation results, assumptions and reviewer approval.

7 6 Production planning and control

Translate accepted demand into feasible jobs, material availability and dated work.

Process Check order and design readiness, explode materials, allocate capacity, schedule operations and monitor completion.

PPC-01 Order readiness check
Verify accepted specification, approved drawing, credit clearance and promised date.

PPC-02 Job creation
Assign a unique production or service job ID linked to the sales order.

PPC-03 Material requirements
Calculate component demand from approved quantities, yield and available stock.

PPC-04 Capacity planning
Reserve work centres, skills and bottleneck time using current calendars.

PPC-05 Operation scheduling
Sequence winding, fabrication, assembly and testing with due dates.

PPC-06 Work in progress visibility
Show job status, completed quantity, waiting reason and responsible department.

PPC-07 Shortage escalation
Assign purchase, substitution or rescheduling actions with approval.

PPC-08 Completion reconciliation
Close only after material, labour, inspection and finished quantity reconcile.

Records and fields job id, order id, design revision, planned quantity, route id, work centre, planned start, planned finish, capacity hours, material shortage, completed quantity, closure date.

Measures Schedule adherence; Work in progress age; Material ready at planned start.

Handoffs Stores reserves stock; procurement resolves shortages; production reports actual progress; QA releases output.

Control Changes to priority must show their effect on other committed orders.

Acceptance A planner can identify the current operation, shortage owner and next promised event for each open job.

7 7 Heater production and winding

Execute approved heater designs with lot traceability, measured geometry and accountable production records.

Process Receive job and materials, verify setup, wind coil, assemble heater, inspect and transfer with recorded actuals.

WND-01 Calculation use *Present in source [L02 L03]*
View calculated wire gauge, length, turns, pitch and selected geometry.

WND-02 Controlled job traveller
Record job, drawing revision, operation sequence and required inspection points.

WND-03 Wire lot issue
Link issued alloy, SWG, supplier lot, length and unused return to the job.

WND-04 Setup verification
Record machine, mandrel, core, pitch setting and first piece approval.

WND-05 Winding progress
Record operator, quantity, start and finish times and interruptions.

WND-06 In process inspection
Record resistance, dimensions and other approved checks with instrument identity.

WND-07 Scrap and rework
Capture rejected quantity, recoverable material, reason and authorised disposition.

WND-08 Production transfer
Move accepted quantities to assembly or QA with traceable acceptance.

Records and fields job id, wire lot, wire swg, wire length issued m, operator id, machine id, setup revision, actual turns, actual pitch mm, actual resistance ohm, scrap quantity, inspection reference.

Measures First pass yield; Wire utilisation; Winding cycle time.

Handoffs Engineering owns design; stores owns issue and return; QA owns acceptance.

Control A calculator result supports setup; manufacturing release needs a reviewed design and inspection evidence.

Acceptance One finished heater traces back to its design revision, wire lot, operator and actual inspection results.

7 8 Mechanical fabrication and assembly

Build frames, ovens, furnaces, conveyors and custom equipment against controlled drawings.

Process Release fabrication drawings, cut and form parts, join and finish, assemble and inspect.

FAB-01 Drawing distribution
Make the current approved fabrication revision available at the work centre.

FAB-02 Material identity
Track sheet, tube and bought out part specifications and supplier evidence.

FAB-03 Cutting and forming records
Record sizes, work orders and material utilisation where needed.

FAB-04 Joining process records
Record applicable joining procedure, person and inspection evidence.

FAB-05 Assembly bill of materials
Verify fitted components against the approved assembly list.

FAB-06 Dimensional inspection
Record critical fits, enclosure dimensions and assembly clearances.

FAB-07 Subcontract tracking
Track material sent out, expected return and inspection on receipt.

FAB-08 Assembly release
Record incomplete items and prevent transfer before accepted closure.

Records and fields fabrication job id, drawing revision, material grade, material lot, part id, assembly id, subcontract reference, critical dimension, inspection result, fitment check, release reference.

Measures Fabrication rework rate; Assembly completion on time; Subcontract turnaround.

Handoffs Engineering releases drawings; procurement buys parts; QA signs inspection; projects receives finished equipment.

Control Approved substitutions and drawing revisions must reach every affected assembly.

Acceptance The assembled equipment matches its bill of materials and has recorded results for all mandatory inspection points.

7 9 Electrical controls and automation

Manage electrical panels, temperature controls, sensors and process automation deliverables.

Process Define control philosophy, approve schematics, build panel, configure software and verify operation.

CTL-01 Electrical drawing control
Version schematics, wiring schedules and panel layout against the equipment job.

CTL-02 Panel component register
Record controller, relay, switchgear and sensor part numbers and ratings.

CTL-03 Software configuration
Version PLC, HMI or controller configuration when used in the project.

CTL-04 Sensor specification
Record thermocouple or RTD type, range, construction and connection requirements.

CTL-05 Wiring inspection
Verify terminations, identification and schematic compliance with recorded results.

CTL-06 Functional sequence test
Test interlocks, alarms and operating modes against an approved procedure.

CTL-07 Configuration backup
Retain released firmware, parameters and recovery instructions with access control.

CTL-08 Panel handover
Provide approved drawings, test records, manuals and spare parts information.

Records and fields panel id, job id, schematic revision, controller model, sensor type, sensor range, configuration version, software backup ref, test procedure, alarm test result, handover revision.

Measures Panel first pass acceptance; Configuration revision defects; Functional test closure time.

Handoffs Engineering approves logic; stores issues components; QA reviews tests; commissioning verifies on site.

Control Safety functions and electrical ratings require the assigned qualified engineering reviewer.

Acceptance The delivered panel has an approved schematic and recoverable configuration matching the tested version.

7 10 Procurement and vendor management

Acquire approved materials and services with traceable price, quality, lead time and receipt controls.

Process Receive requisition, obtain quotations, select approved supplier, approve order, expedite and resolve receipt issues.

BUY-01 Supplier master
Maintain supplier identity, contacts, approved categories and assessment evidence.

BUY-02 Purchase requisition
Link requested material and need date to job, stock policy or approved expense.

BUY-03 Supplier quotation comparison
Compare price, specification, lead time, tax basis and payment terms.

BUY-04 Purchase order approval
Route commitment by value and authority before issue.

BUY-05 Delivery follow up
Track supplier confirmations, partial deliveries and revised dates.

BUY-06 Supplier quality actions
Link rejected receipts to replacement, credit and corrective action.

BUY-07 Subcontract purchase
Specify supplied material, expected work, inspection and return obligations.

BUY-08 Supplier performance
Review on time delivery, accepted quantity and unresolved corrective actions.

Records and fields supplier id, requisition id, job id, rfq id, quotation reference, purchase order id, item id, quantity, unit, promised date, approved price, supplier lot.

Measures Supplier on time delivery; Supplier rejection rate; Purchase cycle time.

Handoffs Planning requests quantities; engineering approves specifications; stores receives; finance matches invoices.

Control Separate requisition, purchase approval, receipt and payment responsibilities where staffing permits.

Acceptance A purchase invoice can be traced to approved order, actual receipt and accepted or rejected quantity.

7 11 Stores and inventory

Protect material identity and maintain quantities by item, location, lot and stock status.

Process Receive, inspect or quarantine, locate, reserve, issue, return, count and reconcile stock.

INV-01 Pipe and core reference lists *Present in source [L03]*
Use hardcoded stock size options to guide heater geometry selection.

INV-02 Item master
Assign item IDs, specifications, units, conversion rules and purchasing/stock categories.

INV-03 Goods receipt
Record supplier delivery, purchase order, quantity and lot evidence.

INV-04 Quarantine control
Keep uninspected, rejected and usable quantities distinct.

INV-05 Material reservation
Allocate quantities to jobs without silently reducing physical stock.

INV-06 Issue and return
Post material transactions with job, person, time and reason.

INV-07 Cycle counting
Record counted quantity, difference and independent adjustment approval.

INV-08 Reorder planning
Use approved lead time, safety stock and demand assumptions for replenishment.

Records and fields item id, warehouse id, bin id, lot id, stock status, receipt id, movement id, job id, quantity, unit, reservation id, count reference, adjustment approval.

Measures Inventory record accuracy; Stockout incidents; Slow moving stock value.

Handoffs Procurement owns order; QA releases quarantined stock; production acknowledges issue; finance reconciles value.

Control Hardcoded size lists describe options; they do not prove available quantities or a warehouse balance.

Acceptance For each lot, opening quantity plus receipts and returns less issues and approved adjustments equals closing stock.

7 12 Quality assurance and inspection

Control product conformance, nonconformities and release using approved plans and retained evidence.

Process Define inspection plan, inspect incoming and in process work, test finished output, resolve deviations and approve release.

QUA-01 Inspection plan
Define characteristics, method, sample basis and acceptance rule by product and revision.

QUA-02 Incoming inspection
Record material checks and supplier evidence before stock release.

QUA-03 In process hold points
Require approved checks before irreversible or downstream operations.

QUA-04 Final test record
Record resistance, insulation, electrical and dimensional tests applicable to the approved product plan.

QUA-05 Nonconformity register
Capture requirement, observed condition, affected lots and immediate containment.

QUA-06 Corrective action
Assign root cause, action, verification and effectiveness review.

QUA-07 Release authorisation
Keep final acceptance distinct from data entry and production completion.

QUA-08 Quality document control
Retain approved procedures, revisions and product release evidence.

Records and fields inspection id, job id, product id, plan revision, characteristic, measured value, unit, acceptance limit, instrument id, ncr id, disposition, release authoriser.

Measures First pass yield; Cost of poor quality; Corrective action overdue.

Handoffs Engineering sets requirements; production contains defects; stores controls status; dispatch checks release.

Control Published test descriptions are company statements; actual procedures, limits and records need company validation.

Acceptance Dispatch is blocked when a mandatory characteristic fails or a required inspection record is missing.

7 13 Calibration and metrology

Manage traceable instrument and equipment results, review and recalibration commitments.

Process Identify instrument and method, confirm reference validity, record readings, evaluate, review and issue controlled report.

CAL-01 Calibration report builder *Present in source [L02 L03]*
Capture customer, instrument, conditions, reference standard, readings and signatory fields.

CAL-02 Automatic error evaluation *Present in source [L03]*
Calculate observed minus standard and compare absolute error with percentage based allowed error.

CAL-03 Report numbering and due date *Present in source [L03]*
Generate sequence and default annual due date while permitting relevant edits.

CAL-04 Calibration register and export *Present in source [L03]*
Search, view, edit, print and export stored reports to CSV.

CAL-05 Linked performance report *Present in source [L03 L04]*
Reuse calibration context and evaluate error in Celsius or percentage with separate register.

CAL-06 Reference standard register
Manage standard identity, certificate, scope, expiry and out of tolerance impact.

CAL-07 Report review and revision
Add independent authorisation, issued version retention and superseded report status.

CAL-08 Uncertainty and decision rule
Record applicable uncertainty, tolerance basis and agreed conformity decision rule.

Records and fields instrument id, report id, report no, customer id, calibration date, due date, standard id, standard certificate, standard validity, method revision, reference value, observed value, error, error unit, tolerance basis, uncertainty, decision rule, reviewer, issued revision.

Measures Calibration turnaround; Overdue instruments; Report amendment rate.

Handoffs Service receives equipment; sales agrees scope; QA reviews method; finance invoices accepted work.

Control Traceability wording on the public website does not establish the company is an accredited laboratory.

Acceptance A report links valid reference evidence, complete readings, the chosen decision rule and its authorised issued revision.

7 14 Projects installation and commissioning

Deliver approved equipment and custom projects through site readiness, installation and customer acceptance.

Process Confirm scope, plan milestones, check site readiness, install, test, resolve punch items and obtain handover.

PRJ-01 Project register
Link equipment, customer, order, project owner, milestones and budget.

PRJ-02 Scope baseline
Record deliverables, exclusions, interfaces and customer responsibilities.

PRJ-03 Site readiness
Confirm utilities, access, foundations and agreed operating conditions.

PRJ-04 Resource scheduling
Assign competent people, travel, tools and required test equipment.

PRJ-05 Installation checklist
Record installation observations and completion against the approved plan.

PRJ-06 Commissioning results
Capture operating tests, settings and deviations using the agreed acceptance procedure.

PRJ-07 Change and punch list
Approve additional scope and record remaining issues before handover.

PRJ-08 Customer acceptance pack
Retain signed acceptance, training, manuals, spares and warranty start reference.

Records and fields project id, order id, site id, equipment serial, project owner, milestone, planned date, actual date, site readiness, punch item, change order, customer acceptance.

Measures Milestone adherence; Project cost variance; Punch list closure time.

Handoffs Sales owns scope; engineering owns drawings; dispatch moves equipment; service receives the installed base.

Control Customer acceptance and commercial closure require explicit evidence for agreed conditions.

Acceptance The project closes only when acceptance requirements and authorised residual actions are recorded.

7 15 Service repair and annual maintenance

Maintain the customer installed base and turn service requests into diagnosed, repaired and accepted work.

Process Log complaint, triage, schedule, diagnose, estimate, authorise repair, test and close with customer acknowledgement.

SRV-01 Installed equipment register
Maintain model, serial, customer site, commissioning and warranty references.

SRV-02 Service ticket
Record reported symptom, urgency, contact, application and safe access needs.

SRV-03 Technician assignment
Match skill, location, availability and tool or spare requirements.

SRV-04 Diagnosis record
Record observed failure, evidence, suspected cause and recommended work.

SRV-05 Repair authorisation
Obtain customer approval or approved warranty decision before chargeable work.

SRV-06 Spares and labour capture
Post consumed parts, travel and actual hours to the service job.

SRV-07 Repair test and closure
Record post repair checks and customer acknowledgement.

SRV-08 Maintenance contracts
Schedule agreed visits, entitlements, renewal dates and excluded work.

Records and fields ticket id, equipment serial, customer site, reported symptom, priority, technician id, diagnosis, repair approval, spare issue, labour hours, warranty decision, closure ack, contract id.

Measures First response time; First visit fix rate; Repeat complaint rate.

Handoffs Sales confirms contract; stores supplies spares; calibration handles reports; finance bills authorised work.

Control Keep warranty entitlement and chargeable scope visible before technician work is invoiced.

Acceptance A closed ticket contains diagnosis, approved scope, actual work, test evidence and customer acknowledgement.

7 16 Dispatch logistics and delivery

Move released goods with the correct documents, packaging, custody and delivery evidence.

Process Confirm release and commercial clearance, pack, prepare documents, dispatch, obtain delivery and resolve discrepancies.

LOG-01 Dispatch readiness
Check quality release, customer address, quantity and required finance clearance.

LOG-02 Packing checklist
Verify product protection, identification, accessories and document pack.

LOG-03 Shipment identity
Assign shipment ID and link every package to job or equipment serial.

LOG-04 Delivery documentation
Prepare delivery note and required transport and tax references after applicability review.

LOG-05 Carrier allocation
Record vehicle or carrier, booking, promised transit and charges.

LOG-06 Dispatch acknowledgement
Record handed over quantity, date and custody evidence.

LOG-07 Proof of delivery
Attach recipient acknowledgement and delivery date to the order.

LOG-08 Returns handling
Log shortages, transit damage, returned goods and credit or replacement decision.

Records and fields shipment id, order id, job id, equipment serial, package count, dispatch date, delivery note, invoice reference, carrier, tracking reference, delivery date, proof of delivery, return id.

Measures On time in full delivery; Transit damage rate; Proof of delivery completion.

Handoffs QA releases; finance checks commercial documents; stores posts stock issue; service receives installed asset details.

Control Dispatch must use the approved destination and match invoice and delivery quantities.

Acceptance A delivered order reconciles released, dispatched and received quantities and preserves supporting documents.

7 17 Finance accounts and treasury

Own the financial record from customer invoice and supplier payment through job margin, cash and period close.

Process Maintain accounts, post approved transactions, reconcile balances, collect/pay, close period and review financial results.

FIN-01 Customer billing
Link invoice lines to order, dispatch or accepted service and approved price.

FIN-02 Receivable control
Track due dates, receipts, credit notes, disputes and collection actions.

FIN-03 Supplier invoice matching
Compare order, accepted receipt and invoice before payment approval.

FIN-04 Payable scheduling
Prioritise due payments with approved bank details and authority.

FIN-05 Bank reconciliation
Match bank transactions to receipts, payments and journals with unresolved item ownership.

FIN-06 Job cost and margin
Reconcile material, labour, subcontract and relevant overhead costs to completed jobs.

FIN-07 Fixed assets and expenses
Record capitalisation decisions, depreciation basis and approved employee expenses.

FIN-08 Period close and reporting
Close reconciled ledgers and produce profit, balance sheet and cash reports for the defined entity.

Records and fields entity id, account code, period, invoice id, order id, tax code, net amount, tax amount, gross amount, due date, receipt id, payment id, bank reference, job cost, asset id.

Measures Receivable days; Gross margin; Close completion time.

Handoffs Sales supplies terms; procurement and stores support matching; HR supplies approved payroll; management reviews results.

Control Tax rates, filing duties and payroll deductions must come from validated applicability and current approved configuration.

Acceptance A period report reconciles to the ledger, and every payment has approved supplier identity and supporting transaction evidence.

7 18 Human resources recruitment and onboarding

Manage approved vacancies and employee entry using controlled documents and retained acceptance.

Process Approve vacancy, screen and interview, approve terms, issue offer, confirm acceptance, appoint and complete onboarding.

HR-01 Offer letter generation *Present in source [L02 L03]*
Fill editable employment terms and produce live A4 preview and print output.

HR-02 Offer confirmation and register *Present in source [L03 L04]*
Confirm offers, maintain searchable records and export them to CSV.

HR-03 Appointment letter generation *Present in source [L02 L03]*
Create the four page appointment format and compensation annexure from entered values.

HR-04 Appointment register *Present in source [L03 L04]*
Issue, search, edit, print and export appointments in a separate register.

HR-05 Vacancy approval
Record business need, approved headcount, role and salary range before recruitment.

HR-06 Candidate and interview tracking
Maintain application, evaluation and selection evidence with restricted access.

HR-07 Offer acceptance and onboarding
Track accepted terms, start date, required documents and joining checklist.

HR-08 Employee master and lifecycle
Assign employee ID and manage role, reporting line, confirmation, transfer and exit.

Records and fields vacancy id, candidate id, offer no, offer date, candidate name, designation, joining date, appointment no, employee id, employee address, gross salary monthly, probation months, notice months, accepted at, hr approver.

Measures Time to fill; Offer acceptance rate; Onboarding completed on time.

Handoffs Management approves vacancy and pay; IT provisions access; payroll receives approved employee and pay details.

Control Generated terms and gross amounts are document values; they do not demonstrate payroll execution or signed acceptance.

Acceptance An employee file links approved vacancy, accepted offer, issued appointment and completed joining requirements.

7 19 Attendance payroll and employee development

Convert approved attendance and compensation into controlled pay and maintain competence and employee lifecycle records.

Process Capture time, approve exceptions, process payroll, review, pay, reconcile and maintain training and performance records.

PAY-01 Attendance and shifts
Record actual attendance, shift calendar and approved corrections.

PAY-02 Leave management
Maintain entitlements, applications, approval, balances and effective dates.

PAY-03 Overtime approval
Link additional work hours to authorisation before pay calculation.

PAY-04 Payroll components
Version effective salary components and deductions by employee and period.

PAY-05 Payroll review and payment
Reconcile gross to net, obtain approval and record payment reference.

PAY-06 Payslip and statutory outputs
Generate outputs from validated jurisdiction and applicability settings.

PAY-07 Competence and training
Track job skills, completed training, assessment and authorisation expiry.

PAY-08 Performance and exit
Record agreed objectives, review outcomes, clearance and final settlement checks.

Records and fields employee id, pay period, attendance date, shift id, leave type, leave balance, overtime hours, salary revision, gross pay, deductions, net pay, pay approval, bank payment reference, training id.

Measures Payroll correction rate; Absence rate; Required training current.

Handoffs HR owns employee master; supervisors approve time; finance approves payments; quality owns technical authorisations.

Control Restrict personal and bank details; use effective dates and retain approved payroll versions.

Acceptance A paid period reconciles approved attendance, pay inputs, deductions, net pay and payment totals without duplicates.

7 20 Administration and facilities

Support daily operations through controlled facility services, records, visitors and shared resources.

Process Receive request, approve cost or access, assign service, verify completion and retain evidence.

ADM-01 Facility request register
Track housekeeping, repairs, office support and responsible service owner.

ADM-02 Visitor and contractor entry
Record host, purpose, approved access and exit according to company policy.

ADM-03 Travel and accommodation
Approve employee travel and link bookings and claims to jobs or cost centres.

ADM-04 Office procurement
Track approved consumables and administrative service purchases.

ADM-05 Document correspondence
Maintain incoming and outgoing official correspondence and response owners.

ADM-06 Asset custody
Assign shared equipment and record issue, return and condition.

ADM-07 Insurance and renewals
Track policy, service agreement and licence renewal dates after owner validation.

ADM-08 Facility service review
Monitor recurring provider obligations, unresolved requests and service cost.

Records and fields request id, facility id, service provider, visitor id, host id, travel request, cost centre, asset custodian, renewal reference, renewal date, completion evidence.

Measures Facility request closure time; Renewals overdue; Administrative cost variance.

Handoffs Procurement contracts services; finance pays verified costs; EHS manages required safety conditions.

Control Access, cost and renewal approvals must be identifiable and time bounded.

Acceptance Each open facility request has an owner, agreed completion date and visible evidence when closed.

7 21 Maintenance and utilities

Keep production machines, tools and utilities available through planned maintenance and accountable repairs.

Process Register asset, assess criticality, plan maintenance, execute work, verify restoration and analyse repeat failures.

MNT-01 Equipment asset register
Identify machines, location, responsible owner and required maintenance information.

MNT-02 Preventive maintenance
Schedule tasks by approved calendar or operating usage.

MNT-03 Breakdown request
Record symptom, equipment, downtime start and production impact.

MNT-04 Maintenance work order
Assign competent staff, spare needs, permits and completion checks.

MNT-05 Spare parts planning
Reserve critical spares and record actual consumption and return.

MNT-06 Restoration verification
Record functional checks and hand back equipment to production.

MNT-07 Utilities monitoring
Capture electricity and other relevant utility meter readings and abnormal consumption.

MNT-08 Failure improvement
Review repeat faults and approve replacement or reliability improvement actions.

Records and fields asset id, criticality, maintenance plan, work order id, failure code, downtime start, downtime end, repair hours, spares consumed, restoration check, meter id, utility quantity.

Measures Unplanned downtime; Preventive maintenance compliance; Repair turnaround time.

Handoffs Production reports failures; stores supplies spares; EHS controls permits; finance records asset spending.

Control Return equipment to use only after required checks and release by the responsible owner.

Acceptance A completed work order includes fault, actual work, consumed spares, downtime and verified return to service.

7 22 Environment health and safety

Manage actual workplace hazards, incidents and environmental obligations through named controls and evidence.

Process Identify hazards, assess risk, authorise work, inspect conditions, investigate events and verify corrective actions.

EHS-01 Hazard and risk register
Record equipment, task, hazard, affected people, controls and review date.

EHS-02 Work authorisation
Use company approved permits and isolation checks for tasks requiring them.

EHS-03 Safety induction
Record employee and contractor induction and relevant training.

EHS-04 Inspection actions
Track workplace inspections, unsafe conditions and corrective action ownership.

EHS-05 Incident and near miss
Capture event, immediate action, investigation and verified closure.

EHS-06 Protective equipment
Record required equipment, issue, inspection and replacement.

EHS-07 Emergency readiness
Maintain responsibilities, equipment checks, drill records and follow up.

EHS-08 Waste and environmental records
Track applicable waste, contractor evidence and resource use against validated obligations.

Records and fields hazard id, task id, risk assessment, control owner, permit id, isolation check, training record, incident id, corrective action, ppe issue, waste record, emergency drill.

Measures Safety actions overdue; Induction completion; Incident investigation closure.

Handoffs Every operating department owns task controls; HR owns training records; facilities and maintenance close physical actions.

Control Confirm which legal and site requirements apply before configuring permits, reporting or retention periods.

Acceptance Work requiring a permit cannot be marked ready while its specified prerequisites are incomplete.

7 23 Information technology data and security

Protect access, source data and recoverability as the existing operations app expands.

Process Provision identity, approve role, control changes, monitor sync and incidents, back up and demonstrate recovery.

ITS-01 Spreadsheet integration *Present in source [L03 L04]*
Read and save the configured wire and document registers through Apps Script.

ITS-02 Local save fallback *Present in source [L03]*
Keep supported document records in browser local storage when shared save fails.

ITS-03 Installable application shell *Present in source [L03 L06]*
Cache same origin application assets and offer supported browser installation.

ITS-04 Identity and access roles
Apply individual accounts and least privilege to every record and write operation.

ITS-05 Audit and document retention
Record actor, action, before and after revision and retention owner.

ITS-06 Reliable synchronisation
Provide visible sync states, idempotent retries, conflict handling and reconciliation.

ITS-07 Backup and restore
Back up shared records and prove restoration against agreed recovery objectives.

ITS-08 Release and incident management
Test changes, version deployments and record service incidents with recovery actions.

Records and fields user id, role id, entity scope, record id, record version, created by, updated by, sync status, request id, audit event id, backup id, restore result, deployment version.

Measures Failed sync unresolved; Restore success; Access reviews overdue.

Handoffs Department owners approve access; finance and HR restrict sensitive records; management owns recovery priorities.

Control Local storage is browser specific; cached app assets do not prove an offline database or safe multiuser conflict handling.

Acceptance An unauthorised user cannot read restricted HR data or submit writes, and an approved restore recovers the tested records.

7 24 Compliance audit and document control

Maintain the approved obligation register, controlled records and independent assurance over department processes.

Process Confirm applicability, assign owner, control document revision, gather evidence, audit exceptions and verify closure.

AUD-01 Obligation register
Record legal, contractual, customer and certification requirements with applicability evidence.

AUD-02 Controlled document library
Version policies, procedures, drawings, formats and approved effective dates.

AUD-03 Record retention schedule
Define retention and disposal from actual legal and business requirements.

AUD-04 Internal audit plan
Schedule independent checks by process, risk and agreed scope.

AUD-05 Evidence and findings
Link each finding to a requirement, record sample and responsible process owner.

AUD-06 Corrective action follow up
Verify completion and effectiveness with evidence separate from action entry.

AUD-07 Contract and certification tracking
Retain signed terms, actual certificates, scope, validity and renewal ownership.

AUD-08 Change impact review
Assess process, regulatory, customer and record consequences before major changes.

Records and fields obligation id, applicability basis, process owner, document id, revision, effective date, approval, retention rule, audit id, finding id, evidence reference, corrective action id, certificate scope.

Measures Audit findings overdue; Controlled document current; Obligation evidence complete.

Handoffs Department leads supply records; management approves policy; qualified advisers validate legal and certification applicability.

Control No registration, accreditation, statutory status or compliance claim is assumed from website wording.

Acceptance An audit finding links a specific requirement and evidence, then closes only after independent verification.

8 End to end workflow model

The six workflows describe the proposed handoffs between functions. They are not claims that every step is already implemented. Each step has an output record, a completion gate and an exception route in the companion workbook. A job changes state only after the defined prerequisite is satisfied.

Workflow	Trigger	Completion evidence
WF01 Enquiry to cash	Customer requests a custom heater	Approved design, material and production trail, accepted shipment, invoice and receipt
WF02 Calibration	Customer requests instrument calibration	Identified item, valid standard, readings, reviewed issued report and custody closure
WF03 Purchase to payment	Approved need for material or service	Approved PO, accepted receipt, matched invoice and reconciled payment
WF04 Employee lifecycle	Approved vacancy	Accepted terms, joining, access, pay and eventual exit clearance
WF05 Service and maintenance	Customer complaint or planned service	Diagnosis, approved work, actuals, test and customer acknowledgement
WF06 Custom equipment project	Accepted project scope	Released designs, build tests, installation, site acceptance and handover pack

Cross department record chain

Customer ID links enquiry, quotation and sales order. The order links approved design revision, job, bill of materials, material movement, production actuals, inspection and equipment serial. Shipment and invoice link back to the accepted order. Service and calibration use the equipment or instrument identity and customer site. Employee IDs link authorised staff to work and approval records without exposing pay data to production users.

Status and exception rules

Use distinct business states such as draft, reviewed, approved, released, in progress, held, accepted, issued, superseded and closed where relevant. Keep the record's business state separate from pending sync, synced, failed or conflict. An apparent successful local save must not be reported as confirmed shared storage.

A missing material lot, expired reference standard, failed test, unapproved cost or customer scope mismatch should create an explicit hold with an owner and next action. Deleting the exception or entering a manual Pass must not substitute for resolution evidence.

8 WF01 Enquiry to cash for custom heater

1 Sales
Capture customer application and requirement. Output: Enquiry and requirement revision. Gate: Application inputs complete. Exception: Return incomplete scope to customer.

2 Engineering
Check feasibility and calculate winding. Output: Design and calculation revision. Gate: Engineer approves design basis. Exception: Revise specification or decline infeasible scope.

3 Estimation
Build cost and issue approved quotation. Output: Estimate and quotation revision. Gate: Margin and discount approved. Exception: Escalate price or scope exception.

4 Sales and Finance
Accept order and check credit or advance. Output: Order and commercial clearance. Gate: PO matches accepted quotation. Exception: Resolve mismatch or hold order.

5 Planning
Release job and material requirement. Output: Job traveller and approved BOM. Gate: Design and material readiness confirmed. Exception: Assign shortage action.

6 Procurement and Stores
Receive, inspect, reserve and issue material. Output: Receipt, lot and issue records. Gate: Only accepted lot issued. Exception: Quarantine or approved substitution.

7 Production
Wind and assemble to approved revision. Output: Operation and consumption records. Gate: First piece and in process checks accepted. Exception: Record rework or scrap.

8 Quality
Perform required final checks. Output: Inspection and release record. Gate: Mandatory tests complete and accepted. Exception: NCR and release hold.

9 Dispatch
Pack and dispatch approved goods. Output: Shipment and delivery note. Gate: QA and finance clearance obtained. Exception: Hold incomplete shipment.

10 Finance
Invoice and collect payment. Output: Invoice receipt and bank reconciliation. Gate: Invoice and delivered quantity reconcile. Exception: Dispute and collection ownership.

11 Service
Maintain installed base and warranty. Output: Equipment and service history. Gate: Acceptance and warranty basis recorded. Exception: Route complaint to service ticket.

Suggested test: walk one representative case through every step, then repeat with one deliberately missing prerequisite. Confirm the normal record chain and the exception owner remain visible.

8 WF02 Calibration request to issued report

1 Service or Sales
Agree scope, method and customer acceptance needs. Output: Service request and quotation. Gate: Instrument and required scope identified. Exception: Clarify unsupported range or method.

2 Calibration
Receive and identify item and condition. Output: Instrument receipt and condition record. Gate: Unique identity and condition accepted. Exception: Hold damaged or unidentified item.

3 Calibration
Confirm standard validity and method competence. Output: Standard certificate and authorisation. Gate: Standard valid for date range and method. Exception: Use approved alternate standard or reschedule.

4 Calibration
Capture conditions and raw readings. Output: Calibration readings. Gate: Mandatory readings complete. Exception: Keep draft status.

5 Calibration
Evaluate error and conformity rule. Output: Calculation and decision record. Gate: Units, tolerance basis and decision rule agreed. Exception: Investigate zero reference and invalid input.

6 Quality reviewer
Review report and authorise release. Output: Reviewed issued revision. Gate: Independent approval and evidence complete. Exception: Correct and retain prior draft history.

7 Calibration
Issue report and communicate next due date. Output: Issued certificate and acknowledgement. Gate: Issued copy matches reviewed version. Exception: Amend through controlled revision.

8 Finance and Service
Bill, return item and schedule agreed recall. Output: Invoice, return acknowledgement and recall. Gate: Commercial and custody records reconcile. Exception: Assign dispute or overdue action.

Suggested test: walk one representative case through every step, then repeat with one deliberately missing prerequisite. Confirm the normal record chain and the exception owner remain visible.

8 WF03 Purchase request to payment

1 Requesting department
Raise defined need and required date. Output: Purchase requisition. Gate: Budget and specification approved. Exception: Return incomplete request.

2 Procurement
Obtain comparable supplier offers. Output: RFQ comparison. Gate: Supplier and technical requirements accepted. Exception: Escalate sole source exception.

3 Approver
Approve commitment and order. Output: Purchase order. Gate: Authority and spend limit met. Exception: Hold for higher approval.

4 Stores and Quality
Receive and inspect delivered material. Output: Receipt and inspection. Gate: Accepted quantity and lot identified. Exception: Quarantine discrepancy.

5 Finance
Match supplier invoice. Output: Three way match evidence. Gate: Order, accepted receipt and invoice reconcile. Exception: Resolve mismatch before payment.

6 Finance approver
Authorise and execute payment. Output: Approved payment and bank reference. Gate: Bank identity checked and payment approved. Exception: Hold suspicious or changed bank details.

7 Finance
Reconcile supplier and bank balances. Output: Reconciliation. Gate: No duplicate invoice or payment. Exception: Investigate reconciling item.

Suggested test: walk one representative case through every step, then repeat with one deliberately missing prerequisite. Confirm the normal record chain and the exception owner remain visible.

8 WF04 Recruitment to employee exit

1 Management and HR
Approve vacancy and terms. Output: Vacancy and pay approval. Gate: Headcount and pay approved. Exception: Revise or reject request.

2 HR and Hiring manager
Evaluate candidate. Output: Interview record. Gate: Selection and reference process complete. Exception: Record rejection or further assessment.

3 HR
Issue offer and obtain acceptance. Output: Offer and acceptance. Gate: Terms approved and acceptance recorded. Exception: Revise offer or close vacancy.

4 HR
Issue appointment and complete joining. Output: Appointment and joining checklist. Gate: Required documents and employee ID complete. Exception: Track outstanding requirements.

5 IT and Department lead
Provision access and training. Output: Access and training record. Gate: Role approved and competence requirements met. Exception: Limit access until approved.

6 Payroll and Finance
Process approved time and pay. Output: Payroll and payment evidence. Gate: Independent review and totals reconcile. Exception: Correct input before pay.

7 HR and Department lead
Manage confirmation, review and changes. Output: Lifecycle and effective pay revisions. Gate: Authorised changes with effective dates. Exception: Escalate unresolved review.

8 HR IT and Finance
Complete exit clearance. Output: Access revocation and final settlement. Gate: Assets, access and balances reconciled. Exception: Retain documented open item owner.

Suggested test: walk one representative case through every step, then repeat with one deliberately missing prerequisite. Confirm the normal record chain and the exception owner remain visible.

8 WF05 Complaint to repair and maintenance renewal

1 Service
Log issue and identify equipment. Output: Ticket and installed asset. Gate: Customer and entitlement identified. Exception: Clarify equipment identity.

2 Service scheduler
Assign technician and plan visit. Output: Visit plan. Gate: Skills, tools and access ready. Exception: Reschedule with customer.

3 Technician
Diagnose and recommend work. Output: Diagnosis and estimate. Gate: Cause evidence and scope recorded. Exception: Escalate technical uncertainty.

4 Customer or Warranty approver
Approve repair scope. Output: Approval or warranty decision. Gate: Chargeable work authorised. Exception: Hold unapproved work.

5 Technician and Stores
Execute repair and post actuals. Output: Labour spares and repair record. Gate: Approved parts and procedure used. Exception: Record extra scope for approval.

6 Quality or Service reviewer
Verify operation and close. Output: Post repair test and acknowledgement. Gate: Required checks and customer acceptance complete. Exception: Reopen unresolved issue.

7 Finance and Service
Bill and schedule future service. Output: Invoice and contract reminder. Gate: Actuals and entitlement reconcile. Exception: Dispute or renewal follow up.

Suggested test: walk one representative case through every step, then repeat with one deliberately missing prerequisite. Confirm the normal record chain and the exception owner remain visible.

8 WF06 Custom machine project to handover

1 Sales and Projects
Confirm project scope and milestones. Output: Scope baseline. Gate: Customer scope and interfaces accepted. Exception: Record exclusion or change request.

2 Engineering
Release mechanical electrical and control designs. Output: Approved design package. Gate: Cross discipline review complete. Exception: Resolve interface conflict.

3 Planning and Procurement
Plan manufacture and long lead items. Output: Schedule and purchase plans. Gate: Capacity and supplier commitments feasible. Exception: Approve schedule revision.

4 Production and Quality
Build assemble and test. Output: Build and factory test evidence. Gate: Agreed factory acceptance complete. Exception: Punch list and repair.

5 Projects
Verify customer site readiness. Output: Readiness checklist. Gate: Required utilities and access ready. Exception: Delay installation with documented reason.

6 Projects and Customer
Install commission and train. Output: Commissioning and training records. Gate: Agreed site acceptance complete. Exception: Approved residual punch list.

7 Projects Finance and Service
Handover bill and activate support. Output: Acceptance pack and equipment record. Gate: Commercial and technical deliverables reconciled. Exception: Assign remaining obligation owner.

Suggested test: walk one representative case through every step, then repeat with one deliberately missing prerequisite. Confirm the normal record chain and the exception owner remain visible.

9 Data architecture and records

The companion dictionary contains 488 entries across current form payloads, reading records, backend index columns and proposed department records. Entries describe fields, not populated transaction records. A field may appear in both a form payload and a spreadsheet index because the representations serve different purposes.

Core record relationships

Record group	Required identity and relationships	Owner
Organisation	Entity, site, department, user and role scope	Management and IT
Commercial	Customer, contact, enquiry, quote revision, order and contract	Sales and Finance
Technical	Product, drawing, BOM, routing, calculation and design revision	Engineering
Supply and stock	Supplier, requisition, PO, receipt, lot, location and movement	Procurement and Stores
Execution	Job, operation, equipment serial, actuals and inspection	Planning Production and Quality
Calibration	Instrument, reference, method, readings, issue and amendment	Calibration and Quality
Financial	Invoice, receipt, payment, account, period and job cost	Finance
People	Candidate, employee, terms, attendance, pay and competence	HR and Payroll

Data rules

Use stable internal IDs independently from editable document numbers. Store units explicitly, distinguish blank from real zero, validate reference keys and retain effective revisions. Money needs currency and a defined tax basis. Dates need local business meaning, and event timestamps need a recorded time zone. Reject invalid or missing prerequisites rather than silently converting them to successful zero values.

Separate the reference geometry list from live inventory. Track inventory by item, lot, location, status and quantity. Link exact design and master data revisions to issued work so later updates do not change the meaning of a historical result.

Current payload and index design

The existing backend uses a searchable sheet row plus JSON payload for each document. That can support the present document register, but a larger system needs validated relationships, revision history, access rules, attachment custody, backup, uniqueness and reconciliation. An approved design should define those requirements before choosing a replacement storage platform.

10 Controls access and recovery

The control register contains 20 requirements with owners, triggers, current evidence and observable acceptance conditions. P0 marks controls to resolve before broader use of sensitive records or controlled issue. This is a proposed implementation priority, not a measured company risk rating.

C01 Individual access and entity scope
Restrict HR/pay and approve each role. Test: Unauthorised user cannot list HR records or save documents.

C02 Reference validity and method scope
Verify standard identity certificate range and validity. Test: Expired or out of scope reference blocks issue.

C03 Conformity decision rule
Agree error unit tolerance basis uncertainty and decision rule. Test: Boundary and zero reference examples produce agreed results.

C04 Server validation of calculations
Revalidate numeric readings, status and permissible payload. Test: Altered client result is rejected or recomputed.

C05 Issued document immutability
Retain issued revision and supersede through controlled amendment. Test: Editing an issued record creates a traceable new revision.

C06 Unique numbering and concurrent use
Allocate controlled number per entity type and period. Test: Two simultaneous issuers receive different valid numbers.

C07 Sync and conflict management
Separate pending synced failed and conflict with idempotent requests. Test: Offline then reconnect preserves both users records once.

C08 Backup and restore
Prove shared record and attachment restoration. Test: Restore sample reconstructs counts IDs payloads and attachments.

C09 Released design revision
Approve calculation drawing BOM and master version. Test: Job cannot use superseded revision without authorised exception.

C10 Material lot status
Separate accepted quarantined rejected and reserved. Test: Quarantined stock cannot be issued to production.

10 Controls continued

C11 Production and quality release
Complete mandatory inspection and resolve failed characteristics. Test: Failed mandatory check blocks shipment release.

C12 Purchase payment segregation
Approve order accepted receipt and invoice match before pay. Test: Receipt or price mismatch remains visible and blocks payment.

C13 Commercial change control
Approve cost price delivery and customer changes. Test: Changed scope updates version and linked commitments.

C14 Employee pay effective dates
Version authorised terms and retain approval evidence. Test: New pay takes effect in approved period only.

C15 Record retention and restricted export
Apply approved retention and protect HR exports and shared files. Test: Role limits and disposal evidence match approved schedule.

C16 Deployment and cache version
Version assets, test upgrade and maintain rollback evidence. Test: Returning user receives compatible app and backend version.

C17 Numeric and geometry boundaries
Reject nonfinite or nonpositive inputs and invalid geometry. Test: Negative zero missing and infeasible geometry have defined outcomes.

C18 CSV export safety
Prevent spreadsheet formula interpretation of user text. Test: Text starting with equals remains literal in exported workbook.

C19 Obligation applicability
Validate registrations, thresholds, due dates and legal requirements. Test: Each configured obligation has an applicability basis and owner.

C20 Customer owned property
Identify custody condition accessory and return evidence. Test: Receipt and return reconcile each instrument and accessory.

Suggested access boundaries

Operators need released work and their assigned execution records. Engineers need design and technical masters. Calibration staff need instrument and reference data; reviewers need issue authority. HR and payroll need restricted personal and pay records. Finance needs commercial and accounting access. Managers need scoped review views. IT administrators require controlled maintenance access and reviewed activity. Confirm actual roles and segregation with the company.

11 Worked examples and verification cases

Illustrative heater calculation using a live wire value

Example inputs are hypothetical: 1200 W, 230 V, heater diameter 14 mm, heater length 500 mm and extra resistance 5 percent. The Nichrome SWG 23 row in the live sheet supplies 3.728 in the ohm column, interpreted by the current code as ohms per metre. These inputs do not represent a customer order. [L03 G01]

Quantity	Calculation	Result
Base resistance	230 x 230 / 1200	44.0833 ohm
Adjusted resistance	44.0833 x 1.05	46.2875 ohm
Candidate wire length	46.2875 / 3.728	12.4162 m
Heater area	pi x 14 x 500 / 100	219.9115 cm2
Watt density	1200 / 219.9115	5.4567 W/cm2
Insulation rule	Watt density below 8	1.5 mm in current app logic

This example verifies electrical arithmetic only. Final gauge selection still requires the full geometry, pitch, thermal and application checks. The supplied wire master does not include an independently verified material certificate or temperature correction basis.

Illustrative calibration and performance boundaries

Case	Inputs	Expected existing logic
Calibration within limit	Standard 100; device 100.8; tolerance 1 percent	Error 0.8; allowed 1; Pass
Calibration outside limit	Standard 200; device 202.5; tolerance 1 percent	Error 2.5; allowed 2; Fail row and Review overall
Calibration zero input	Standard 0; device 0.1; tolerance 1 percent	Allowed 0; Fail under current rule
Performance percentage zero	Reference 0 in percentage mode	No calculable percentage; save needs completed valid results
Performance absolute zero	Reference 0; observed 0.2; accuracy 0.5 Celsius	Error 0.2 Celsius; Pass

The workbook contains the heater example as editable inputs and formulas. It also includes example input cells for baseline measurement without invented actual savings.

12 Performance measurement and benefits

The 72 proposed metrics cover all 24 functions. The register defines the population, formula, period and cautions. Actual baselines and targets remain Not supplied. Establish them from reconciled company data before reporting improvement.

How to establish the baseline

Choose a documented period and comparable product or service cohort. Record counts, actual labour time, elapsed turnaround, error or rework events and customer completion dates. Define start and finish timestamps, planned hold treatment and missing record rules. Preserve the baseline extracts and reconcile totals with the owning register.

For the calculator, measure time from accepted input to reviewed design output, separately from prototype validation. For calibration, separate reading acquisition time, report preparation and independent review. For letters, separate document preparation from recruitment and onboarding lead time. A faster screen interaction is not by itself a shorter complete business cycle.

Benefits calculation

Hours released = eligible transaction count multiplied by verified minutes saved per transaction divided by 60. A cash saving requires a documented reduction in paid overtime, outsourced cost or another actual expense. Capacity released can be reported as capacity, with the assumptions and opportunity to use it stated. Do not equate all saved minutes with cash.

Quality benefits should use actual avoided rework or claim cost, including the same cost basis before and after. Delivery improvement should compare matching due date cohorts. Report changes in product mix, staffing, workload and data coverage so the effect of automation is not overstated.

Measurement ownership

Process owners validate event definitions and record completeness. Finance validates cost and recognised financial effects. Management approves targets and review cadence. IT validates data refresh and duplicate prevention. Each metric should have an owner who can explain its calculation and missing data.

13 Implementation sequence and acceptance

The phases below are dependency based. No dates, project cost or staffing promise is assumed. Agree duration and resources after validating process variants, source data and deployment constraints.

PH0 Validate scope and evidence
Owner: Management and department leads. Work: Confirm entities, owners, process variants and baseline; fill data requests. Exit evidence: Approved scope, owners and evidence register.

PH1 Protect existing records
Owner: IT and record owners. Work: Individual access, server validation, versioning, unique numbers and restore. Exit evidence: Access, concurrency, tamper and restore acceptance pass.

PH2 Control engineering and quality
Owner: Engineering Quality and Calibration. Work: Approve design history, standards, decision rules and controlled report release. Exit evidence: Representative design and calibration journeys pass.

PH3 Connect demand material and production
Owner: Sales Planning Procurement Stores Production. Work: Create order job BOM material movements and production inspection links. Exit evidence: One heater order reconciles from requirement to delivery.

PH4 Connect finance projects and service
Owner: Finance Projects Service Dispatch. Work: Link accepted output to invoices cost receipts warranty and installed base. Exit evidence: Invoice and job cost reconcile; project/service acceptance captured.

PH5 Complete employee and support processes
Owner: HR Payroll Admin Maintenance EHS Compliance. Work: Approved employee lifecycle, pay, assets, maintenance and obligation workflows. Exit evidence: Restricted payroll trial and applicable support workflows pass.

PH6 Measure and improve
Owner: Management and process owners. Work: Compare approved baseline with complete operating periods. Exit evidence: Measured benefits reconciled and improvement owners named.

Acceptance before expansion

Run one custom heater order, one calibration issue, one purchase payment, one employee joining, one service repair and one custom equipment project through the connected record chain. Repeat with missing approvals, failed tests, expired standards, a changed design, a partial delivery and an offline save. The correct owner and hold must remain visible.

For technical release, test simultaneous numbering, altered client payloads, access restrictions, browser upgrades, backup restoration and reconnect conflicts. Reconcile counts, IDs, revisions, values and custody references. An export succeeding or a dashboard showing a total does not prove the workflow is complete.

14 Company data still required

The following records are needed to validate the organisation, complete the actual company dataset and quantify outcomes. The workbook provides 26 requests with an editable owner, due date, evidence reference and collection status. Actual personal and pay records should be restricted to their authorised owners.

REQ-001 Legal entity and group structure Registration documents, sites, authorised signatories and entity relationships. Owner: Management.

REQ-002 Actual organisation chart Departments, reporting lines, role owners and delegation limits. Owner: Management.

REQ-003 Financial statements and trial balances Latest approved periods, ledger mapping and entity scope. Owner: Finance.

REQ-004 Customer and order registers Customer IDs, enquiries, quotes, orders, dates, values and status history. Owner: Sales.

REQ-005 Approved specifications and design history Drawings, design calculations, BOMs, revisions and engineering approvals. Owner: Engineering.

REQ-006 Wire master and material evidence Actual wire rows, alloy certification, resistance units and master revisions. Owner: Engineering.

REQ-007 Jobs routes and capacity Production schedule, work centre calendars, routing and actual progress. Owner: Planning.

REQ-008 Production actuals Operation time, output, rework, scrap and job material consumption. Owner: Production.

REQ-009 Inventory and movement history Opening balances, receipts, issues, returns, lots, locations and count results. Owner: Stores.

REQ-010 Suppliers and purchasing Vendor approvals, requisitions, quotes, POs, receipts and invoice links. Owner: Procurement.

REQ-011 Inspection and nonconformity records Plans, measured results, release evidence, NCRs and corrective actions. Owner: Quality.

REQ-012 Actual report and standard records Registers, raw readings, issued revisions, reference certificates, scope and validity. Owner: Calibration.

REQ-013 Project and acceptance records Scope, milestones, budgets, actuals, commissioning and customer signoff. Owner: Projects.

14 Company data continued

REQ-014 Installed base service and contracts Equipment serials, tickets, diagnosis, parts, labour, warranty and AMC terms. Owner: Service.

REQ-015 Shipment and delivery records Delivery notes, invoice links, carrier, dates, POD and damage claims. Owner: Dispatch.

REQ-016 Cost and cash records Rates, job costing, invoices, ageing, receipts, payments and bank reconciliations. Owner: Finance.

REQ-017 Restricted employee lifecycle records Approved vacancies, accepted offers, appointments, joins, changes and exits. Owner: HR.

REQ-018 Restricted attendance and pay evidence Time, leave, overtime approvals, pay components, deductions and payments. Owner: Payroll.

REQ-019 Assets and downtime Asset list, preventive plans, breakdowns, repairs, spares and meter readings. Owner: Maintenance.

REQ-020 Facility services and renewals Supplier contracts, assigned assets, travel and renewal calendar. Owner: Administration.

REQ-021 Applicable safety and environment records Risk assessments, permits, training, incidents, inspections and waste records. Owner: EHS.

REQ-022 Deployment access and storage evidence Hosting version, Apps Script deployment access, user roles and record counts. Owner: IT.

REQ-023 Backup restore and sync evidence Backup jobs, restore results, failed sync history and conflict resolutions. Owner: IT.

REQ-024 Registration certification and obligation evidence Actual certificates, accreditation scope, applicability register and audit results. Owner: Compliance.

REQ-025 Catalogue completeness Current full catalogue, discontinued items, SKU IDs and missing web pages. Owner: Marketing.

REQ-026 Baseline and target definitions Measured volumes, elapsed time, error rates, date window and approved targets. Owner: All department owners.

Completion standard

An evidence complete company case study requires the actual structure and process owners, complete current product master, relevant internal registers and measured outcomes. This package provides the full proposed department coverage and all retrieved wire data while keeping outstanding factual records explicit. No claim is made that inaccessible or absent internal company data has been obtained.

15 Sources and audit trail

Public websites were reviewed on 15 October 2026. Company published descriptions are attributed to the company and do not establish independent verification. Local source observations describe the files reviewed on the same date. The live spreadsheet source is a connected read of the two visible tabs, not a modification.

S01 Company business profile Public company statement.
https://electrotechind.in/
Heating and packaging activities; not audited performance.

S02 Electrotech about Public company statement.
https://electrotechind.in/about/
Manufacturing and testing statements; no independent certification verification.

S03 Electrotech services Public company statement.
https://electrotechind.in/our-services/
Repair and calibration; website says servicing since 1995.

S04 Electrotech contact Public company statement.
https://electrotechind.in/contact-us/
Baikampady address, public business phone and email.

S05 Packaging machinery catalogue Public catalogue.
https://electrotechind.in/product-category/packaging-machineries/
12 of 14 listed items accessible; page two unavailable.

S06 Plastic processing catalogue Public catalogue.
https://electrotechind.in/product-category/plastic-processing-equipment/
Three catalogue entries.

S07 Heating equipment catalogue Public catalogue.
https://electrotechind.in/product-category/heating-equipment/
Seven catalogue entries.

S08 Controllers and panels catalogue Public catalogue.
https://electrotechind.in/product-category/temperature-controllers-panel-box/
Three catalogue entries.

S09 Sensors catalogue Public catalogue.
https://electrotechind.in/product-category/thermocouples-sensors/
One category entry; not a complete sensor SKU list.

S10 Packaging materials catalogue Public catalogue.
https://electrotechind.in/product-category/packaging-materials/
Three catalogue entries.

S11 Customized machines catalogue Public catalogue.
https://electrotechind.in/product-category/customized-machines/
Six catalogue entries.

S12 Electro Heat Systems profile Public company statement.
https://electroheatsystems.in/
Reciprocal website links establish association, not legal ownership.

S13 Electro Heat Systems about Public company statement.
https://electroheatsystems.in/about/
Manufacturing since 1998 and listed test types are company claims.

S14 Electro Heat Systems catalogue Public catalogue.
https://electroheatsystems.in/products/
15 catalogue entries; no prices or capacity verified.

L01 Project overview Local source review.
readme.md
Source inventory; not a live deployment test.

L02 Application interface Local source review.
index.html
Navigation, form fields and defaults.

L03 Application logic Local source review.
script.js
Calculation rules, registers, exports and local storage.

L04 Spreadsheet backend Local source review.
google-sheets/Code.gs
Routes, table headers, validation and locking.

L05 Backend setup guide Local source review.
google-sheets/README.md
Deployment instructions and access limitations.

L06 Installable application Local source review.
service-worker.js and manifest.webmanifest
App shell caching and install settings.

A01 Operating model recommendations Analysis.
Case study department and workflow design
Proposed functions; actual organisation and policies require validation.

G01 Live wire master Connected Google Sheets read.
https://docs.google.com/spreadsheets/d/1fuzplqthrPBaIbyuaGIOg_n1Z8XnSSdimjJjArl76So/edit
Read both visible tabs A1:F1000 on 15 October 2026; 23 Nichrome and 9 Kanthal rows; no document register tabs visible.

Excluded identity match: similarly named IT, electrical contracting and overseas businesses were not used as evidence for this Baikampady company. The associated Electro Heat Systems website was included because the two company websites link to each other; ownership remains a data request.

Appendix All retrieved wire records

Values below reproduce the connected sheet rows. Thickness is used as millimetres and ohm is used as ohms per metre by the application. The wattage bounds are inclusive. No values have been corrected. Engineering must validate disputed rows before a controlled master revision. [G01 L03]

Material	ID	SWG	Thickness	ohm	minw W	maxw W
nichrome	1	22	0.711	2.76	1300	1599
nichrome	2	23	0.61	3.728	1200	1299
nichrome	3	24	0.559	4.536	900	1199
nichrome	4	25	0.508	5.31	850	1049
nichrome	5	26	0.457	6.802	800	899
nichrome	6	28	0.376	9.793	600	799
nichrome	7	30	0.305	13.744	400	599
nichrome	8	32	0.274	18.32	300	399
nichrome	9	32	0.274	18.32	300	399
nichrome	10	34	0.234	25.9	250	299
nichrome	11	35	0.213	31.596	225	274
nichrome	12	36	0.193	36.06	200	249
nichrome	13	36	0.193	36.06	200	249
nichrome	14	37	0.173	45.82	175	199
nichrome	15	38	0.152	62.88	150	199
nichrome	16	38	0.152	59.408	150	199
nichrome	17	39	0.132	78.4	133	174
nichrome	18	40	0.122	93.2	125	149
nichrome	19	40	0.122	94	125	149
nichrome	20	42	0.102	139.2	100	124
nichrome	21	42	0.102	139.2	100	124
nichrome	22	44	0.081	214.66	10	99
nichrome	23	44	0.081	214.66	10	99
kanthal	1	20	0.914	1.668	1900	2099
kanthal	2	21	0.813	2.103	1600	1899

Material	ID	SWG	Thickness	ohm	minw W	maxw W
kanthal	3	22	0.711	2.72	1300	1590
kanthal	4	23	0.61	3.691	1200	1299
kanthal	5	24	0.559	4.55	900	1199
kanthal	6	26	0.508	6.755	800	899
kanthal	7	28	0.376	9.84	600	799
kanthal	8	30	0.315	14.417	400	599
kanthal	9	32	0.274	17.83	300	399

The spreadsheet register also contains the complete hardcoded geometry reference options. Those values have no on hand quantity and are not an inventory count.